Mind the Gap: Reconstructing Missing Values

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Introduction

Alex's story. At the Aurora Science Museum, a new exhibit will recreate the indoor environment of a historic house so visitors can experience life *two to three centuries ago*. The most critical piece is temperature: the team wants the room to follow the climate of the 1700s–1800s. When Alex retrieves nineteenth-century temperature records, the series is riddled with gaps—missing months and even whole years. To ensure authenticity, Alex plans to try four cleaning approaches on the same data slice such as listwise deletion, climatological within-month imputation (mean/median), moving-average smoothing, and Kalman state-space smoothing, and then select the method that best restores the past. **Now, please step in and help Alex complete this task.**

Figure 1: **Reconstructed Israelite house (Monarchy period), Eretz Israel Museum, Tel Aviv.**

Source & license: Talmoryair, *"A reconstructed Israelite house, Monarchy period"* (2006), Wikimedia Commons.

Learning Goals

After completing this lab you will be able to:

- Diagnose missingness patterns over time and by calendar month.

- Explain and compare four methods: *Drop Missing*, *Monthly Mean*, *Monthly Median*, *Kalman Smoothing (season-aware)*, and *Moving Average*.
- Use interactive widgets to contrast methods and parameters, and observe trend continuity, seasonal preservation, and lag.
- Write a short, justified recommendation for Alex under specific data constraints.

1 Dataset Introduction

The dataset used in this project is from the **Berkeley Earth Surface Temperature Study**, available on Kaggle as *GlobalLandTemperaturesByCity.csv*. It contains monthly average surface air temperatures (in °C) for thousands of cities worldwide. In our analysis, we filter to `City = ‘‘Aurora’’` and focus on the variable `AverageTemperature`, which records the monthly mean temperature for that location.

2 Data & Missingness at a Glance

Hands-on Task

Run **Notebook Cell 3** to load Aurora. Then run **Notebook Cell 4** to visualize missingness: (i) timeline of missing vs observed; (ii) missing counts by calendar month.

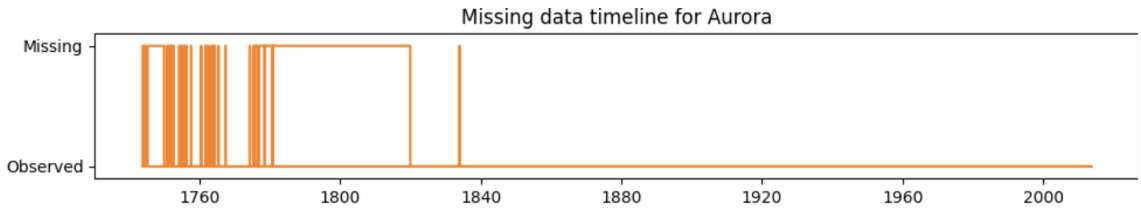
Missingness Diagnostics for Aurora

Dataset profile (Aurora, monthly)

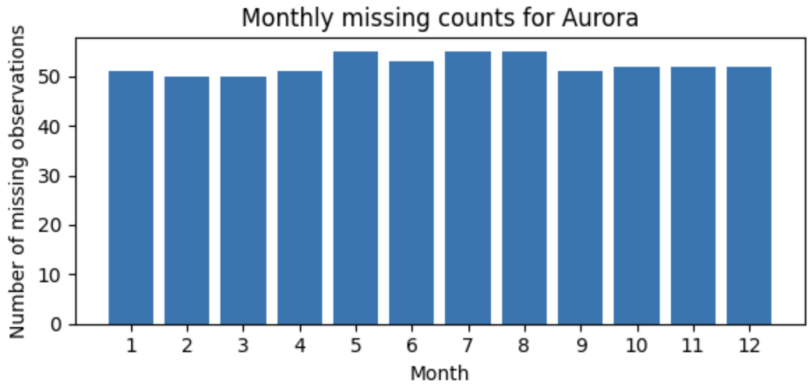
- **Source & unit:** Berkeley Earth city series, variable `AverageTemperature` in °C, aggregated to **monthly** means.
- **Temporal coverage:** Long historical span with sparse early decades and dense modern coverage (exact years depend on the city filter).
- **Seasonality:** Strong 12-month cycle (typical mid-latitude pattern); methods should preserve this cycle.

What the two diagnostics show (Cell Notebook Cell 4)

- **Timeline (Missing vs. Observed):** A *large contiguous block of missing values* in the early record, followed by *near-continuous observations* in later decades. This means any method that simply drops NA will effectively remove the entire early span.
- **By-month counts:** Bars are *roughly flat across months* (no strong January–December bias). That is good news for *within-month* (climatological) imputation: each calendar month has comparable support.



(a) Missing data timeline for Aurora (1 = missing, 0 = observed).



(b) Monthly missing counts for Aurora.

Figure 2: Diagnostics of missingness patterns in Aurora's monthly temperature series.

Hands-on Task

[Interpreting the Diagnostics] Look at the two diagnostics you just produced (timeline and by-month counts).
Step 1. Describe in one or two sentences what you notice in the current data:

- Long gaps in the early record?
- Nearly complete coverage in later years?
- Any obvious month-to-month imbalance?

Step 2. Which of the following best describes what a "good" temperature time series should look like?

1. Continuous record with few or no missing months.
2. Clear seasonal wave (peaks in summer, troughs in winter).
3. Balanced support across months (no single month missing almost entirely).

Circle or note your choices. This will prepare you to understand why different cleaning methods matter.

3 Define the Four Methods (Cell 5)

Open **Notebook Cell 5**. Below are the four cleaning methods in **simple terms**. For each method we explain: *what it does*, *when to use it*, *what to watch out for*.

1. Drop Missing (listwise deletion)

What it does. Simply remove rows that contain NA (missing) values. **When to use.** Missing values are very rare in your chosen window (e.g., $< 5\%$) and you do not need perfect continuity. **Watch out.** Long gaps break the time line and can distort summaries of early decades.

2. Monthly Mean / Monthly Median (same-month fill)

What it does. If January is missing, fill it with the *typical* January value computed from the other years. **Why it helps.** It respects the 12-month seasonal cycle (winter is cold, summer is warm). **When to use.** You have enough data in each calendar month (e.g., at least a handful of observed Januaries, Februaries, ...). **Watch out.** It can *shrink extremes* (filled values look more "average") and ignores year-to-year changes inside the same month.

Tiny formula (optional): for a missing month t with calendar month $\mathrm{mon}(t)$,

$$\hat{y}_t = \begin{cases} \text{mean of all observed values in month } \mathrm{mon}(t), & \text{(Monthly mean)} \\ \text{median of all observed values in month } \mathrm{mon}(t), & \text{(Monthly median)} \end{cases}$$

and leave observed y_t unchanged.

3. Moving Average (window smoothing, fill only where NA)

What it does. For a missing month, average the nearby months in a small sliding window (e.g., ± 6 months). **Why it helps.** Good at bridging *short* gaps using local information. **When to use.** There are observed neighbors near the gap; choose a modest window (e.g., 6–12 months). **Watch out.** Larger windows cause *lag* (peaks/troughs shift and shrink). It struggles in the middle of very *long* gaps.

Tiny formula for a missing t,

$$\hat{y}_t = \text{average of available neighbors in a centered } w\text{-month window around } t.$$

4. Season-aware Kalman Smoothing

Idea in one line. Keep the year-within-month pattern (season), smooth the noisy part, and fill the gaps.

What we do (3 quick steps).

1. **Remove the season.** For each calendar month (Jan, Feb, ...), compute a typical value ("climatology") and subtract it. This leaves us with a de-seasoned series of *anomalies*.

2. **Smooth the anomalies.** Run a *Kalman smoother* on the de-seasoned series. It smooths noise and naturally bridges missing points.

3. **Add the season back.** Put the monthly climatology back to reconstruct the final series with clear winter–summer shape.

What is Kalman smoothing. Assume there is a slowly changing, unobserved "true" signal x_t and what we see is that signal plus measurement noise. Kalman *filtering* uses past data to estimate x_t; Kalman *smoothing* goes further and uses both past *and future* data to refine every time point, so it is steadier and handles gaps well.

Tiny model.

$$x_t = x_{t-1} + w_t \quad \text{(slow drift)}, \qquad \text{anomaly}_t = x_t + v_t \quad \text{(noisy observation after removing season)},$$

where w_t and v_t are small random shocks. We smooth the anomalies to get $\hat{x}_t$ and then set

$$\hat{y}_t \;=\; \hat{x}_t \;+\; \text{season}(t).$$

Why/when this helps.

- **Preserves the 12-month cycle:** winter lows and summer highs remain visible.
- **Bridges long gaps:** unlike simple moving averages, it can span the middle of long missing stretches.
- **Use when:** seasonality is strong and you can estimate a stable monthly climatology from a long time span.

Tuning (what to watch). The smoother balances "follow the data" vs. "be smooth." If you set it too smooth, peaks/troughs get flattened; if you set it too responsive, noise leaks through. Also, if the monthly climatology is poor, very long gaps may still look too flat.

4 Method A — Drop Missing (Cell 6)

Hands-on Task

Run **Notebook Cell 6**. Use the *Start year* and *Window length* sliders to choose a 10-year slice that contains missing values. Observe the overlaid plot: original (with gaps) vs. the "dropped" series.

Think:

- How many points did you drop in this window? Does the result still cover all seasons without large holes?
- Would Alex trust a budget computed from a series with broken continuity?

What we notice

- **Continuity loss.** In decades with long gaps, the kept series becomes fragmented and jagged. The 12-month seasonal wave is no longer continuous; entire months/years can vanish from the slice.

- **Seasonal distortion.** With missing months removed, peaks and troughs shift or disappear, making trend and seasonality hard to interpret.
- **Selection bias.** If missingness clusters in certain months or eras, the remaining points over-represent other periods and skew summaries.
- **Small-n volatility.** After dropping many points, averages and extremes bounce around and become unstable.

Conclusion for Method A

Use "Drop Missing" only when the missing rate is very low and well dispersed (e.g., $<5\%$). Otherwise, prefer an imputation method that preserves continuity and the 12-month seasonal pattern; dropping will likely hide the cycle and degrade any subsequent analysis.
Since "Drop Missing" may distort continuity and seasonality, a next question is: *Can we fill gaps with values that reflect typical seasonal behavior?* This brings us to **Method B—using monthly means or medians** for imputation.

5 Method B — Monthly Mean / Monthly Median (Cell 7)

Hands-on Task

Run **Notebook Cell 7**. Switch between *Monthly mean* and *Monthly median*. Keep the same 10-year window for fair comparison. Watch where imputed points are highlighted.

Think:

- Does the seasonal cycle (peaks in summer, troughs in winter) remain in the cleaned series?
- How much are extremes damped? Is median noticeably flatter than mean?
- In months with very few observations, do you trust the monthly statistic?

What we notice

- **Season preserved.** The cleaned line keeps a clear 12-month wave; imputed points (the blue dots) sit plausibly on that wave.
- **Mean vs. Median.** *Monthly mean* retains slightly larger amplitudes but is sensitive to outliers; *Monthly median* looks a bit flatter and is more robust when a month has a few extreme values.
- **Variance damping.** Extremes (very hot/cold months) are pulled toward the center; long gaps may look a touch "too regular."

- **Data-poor months.** If the console warns many months have few samples (e.g. < 6), month-level statistics can be noisy and the fill may wobble or shift.
- **Trend blindness.** Because we reuse the same month's statistic across years, slow inter-annual drifts (warming/cooling) inside the window are not reconstructed.

Conclusion for Method B

Pros.

- Simple, fast, and **leverages seasonality**; outputs are easy to explain.
- **Median** option is robust to outliers and small spikes.
- Works well when each calendar month has a **sufficient number of observations**.

Cons.

- **Damps extremes** and may slightly reduce variability.
- **Ignores inter-annual trend** within the window; every January is treated alike.
- When month-level data are **very sparse**, the monthly statistic becomes unstable and the imputation can bias the curve.

Conclusion (Method B). Monthly mean/median is a strong baseline for seasonal climate series: it preserves the annual rhythm and produces visually coherent fills. Prefer *median* when outliers exist or month-level counts are modest; prefer *mean* when you want a bit more amplitude and have enough data per month. If your window shows long early gaps or strong year-to-year trend, consider a season-aware model (e.g., Kalman) in later steps.
While monthly means and medians preserve the seasonal cycle, they may smooth away year-to-year variability and damp extremes. To better capture local patterns around each gap, we can instead rely on nearby observations through a moving average. This brings us to **Method C — using sliding windows to impute missing values.**

Hands-on Task

[Extension — Beyond Monthly Means?] After trying *Monthly Mean/Median*, ask yourself:
What if we grouped data by **two-month intervals** (bi-monthly) instead of single months?

- Would this make the imputed series smoother, or would it blur important seasonal peaks and troughs?
- How might this affect winter vs. summer comparisons?

Discuss your intuition briefly. You do not need to run code here—just think about the trade-offs.

6 Method C — Moving Average (Cell 8)

Hands-on Task

Run **Notebook Cell 8**. Adjust the *MA window (months)* slider. Keep the same 10-year window. Note: if a gap is longer than the effective neighborhood, the mid-gap points may remain unfilled.

Think:

- How does increasing the window change lag and amplitude?
- Does moving average fill short gaps well but struggle in the middle of long gaps?
- Would the induced lag mislead a cost curve that tracks seasonal peaks?

What we notice

- **Small window (e.g., 3–6 months):** the cleaned line follows the original closely and reduces noise; short gaps are filled plausibly using nearby months.
- **Large window (e.g., $\geq$ 12 months):** peaks/troughs are *delayed* (phase lag) and amplitudes are *shrunk* (over-smoothing). With a 36-month window, the annual cycle can be averaged away toward a flat baseline.
- **Gap length matters:** short runs of missing values are bridged; in the *middle of long gaps* the curve becomes flat/linear because faraway neighbors dominate. If the gap exceeds the effective neighborhood, mid-gap values are poorly estimated.
- **Boundary effect:** at the *start/end* of the window the average is one-sided, causing extra bias. (Centering the window helps but does not remove lag entirely.)
- **Sample size vs. window:** if `MA window` > available non-missing points (or > the shortest continuous run), results become unstable. Prefer w that is clearly smaller than the length of local non-missing segments.

Conclusion for Method C

Moving average is a **good local patcher for short gaps** and light denoising. Choose a *modest window* (often 3–6 months) to preserve seasonality and limit lag. Avoid large windows near 12 months if you need the annual cycle, and **do not rely on MA for long gaps**—it will either lag heavily or flatten the signal. When long gaps must be bridged *and* seasonal shape kept, prefer a season-aware method (e.g., Kalman with a monthly baseline).

While moving averages work well for short gaps, they struggle when the missing span is long: values in the middle of the gap cannot be anchored by nearby observations, and the series tends to flatten. To handle these cases more intelligently, we can combine a seasonal baseline with a state-space model that smooths anomalies. This brings us to **Method D — season-aware Kalman smoothing.**

7 Method D — Season-aware Kalman (Cell 9)

Hands-on Task

Run **Notebook Cell 9**. Parameters are fixed for simplicity (process noise q, observation noise r, EM iterations). The method uses the full series to compute a monthly climatology, smooths anomalies, then adds season back. Use the same 10-year window.

Think:

- Does the method preserve the within-year shape while bridging long gaps more naturally than moving average?
- Where does it over-smooth or under-smooth? If you could tune q/r, what would you change?

What we notice

- **Season preserved:** peaks and troughs recur at the right months because the method removes the monthly climatology, smooths the residuals, and then restores seasonality. The cleaned curve aligns phase-wise with the original.
- **Long-gap behavior:** in stretches with many missing months, the line does not collapse to a flat segment; it continues a plausible seasonal oscillation guided by the climatology, unlike moving average which may flatten mid-gap.
- **Smoothness vs. fidelity:** with the fixed $q = 0.05, r = 1.0$, the curve is smooth but still responsive. You may still see slight damping of extremes (very hot/cold months are pulled toward typical values).
- **Edge stability:** at window boundaries the method remains stable because it leverages the full-series climatology rather than only local neighbors.
- **Failure modes to watch:** if the climatology is poorly estimated (e.g., very sparse early decades) or if the true period differs from 12 months, residuals can be biased and the reconstruction may look too regular.

Conclusion for Method D

Season-aware Kalman offers the best balance when you need to **bridge long gaps and retain the annual cycle**. With a reasonable q/r choice, it yields a smooth, phase-correct curve that keeps seasonal structure. If extremes matter (heat waves/cold snaps), consider slightly larger q (more responsiveness) or complement with a local method for very short gaps.

8 All-in-one Comparison (Cell 10)

Hands-on Task

Run **Notebook Cell 10**. Switch methods in the dropdown and (for Moving Average) change the window. Use the same start year and 10-year window to compare fairly. Pay attention to the on-screen sample-size warnings (e.g., months with sparse data; MA window larger than available samples).

Think:

- Which method best balances continuity and seasonal fidelity for Alex's exhibit?
- Where do you see lag, and where do you see loss of amplitude?
- If the exhibit focuses on visitor intuition about "how winter felt," which errors are more tolerable?

What we notice

- **Continuity vs. data loss.** *Drop Missing* keeps only observed months—clean but discontinuous. It works only when the chosen window happens to have dense coverage.
- **Seasonal fidelity.** *Monthly Mean/Median* preserve the 12-month rhythm well. Median is slightly flatter (robust to outliers); mean may track extremes a bit more but can be pulled by outliers.
- **Lag behavior.** *Moving Average* introduces phase lag that grows with the window; it handles short gaps nicely but fails in the middle of long gaps (little to no information).
- **Long-gap bridging.** *Season-aware Kalman* is the only method that plausibly spans long early gaps while keeping month-of-year timing, because it smooths anomalies around a climatological seasonal baseline.
- **Sensitivity to sample size.** When per-month counts are very low, monthly statistics become unstable; when the MA window exceeds available neighbors, the curve degrades; Kalman depends on a reasonable climatology estimated from a longer span.

Conclusion for the comparison

For the exhibit's goal—helping visitors *feel* historical winters—methods that retain the seasonal cycle with minimal artifacts are preferable. In most windows with appreciable gaps, **season-aware Kalman** provides the best overall balance (continuity + seasonality). If gaps are short and local fidelity is key, **Moving Average** with a small window is acceptable but be mindful of lag. With moderate missingness and reasonable monthly counts, **Monthly Median** is a solid, robust baseline. **Drop Missing** should be reserved for windows with near-complete coverage.

9 Conclusion & Reflection

Write a short recommendation to Alex:

- **Your choice:** Which method would you use for the exhibit's 10-year slice, and why?
- **Trade-offs:** What do you gain and lose (continuity, seasonal shape, extremes, lag)?
- **Rules of thumb (for future data):**
  - Missing $< 5\%$ and continuity not essential $\rightarrow$ Drop Missing.
  - Strong seasonality with adequate monthly support $\rightarrow$ Monthly Mean/Median (median if outliers).
  - Short gaps with neighbors, need a quick fix $\rightarrow$ Moving Average (window $<$ neighbor count).
  - Long gaps, want seasonal shape preserved $\rightarrow$ Season-aware Kalman (use a longer reference period for climatology).

Appendix: What to Run in the Notebook

- **Notebook Cell 1**: Install minimal dependencies.
- **Notebook Cell 2**: Test imports.
- **Notebook Cell 3**: Download dataset; filter `City="Aurora"`.
- **Notebook Cell 4**: Missingness timeline and by-month bar plot.
- **Notebook Cell 5**: Define the four cleaning methods.
- **Notebook Cell 6**: Drop Missing (interactive).
- **Notebook Cell 7**: Monthly Mean / Monthly Median (interactive).
- **Notebook Cell 8**: Moving Average (interactive).
- **Notebook Cell 9**: Season-aware Kalman (fixed params; interactive).
- **Notebook Cell 10**: All-in-one comparison (interactive).